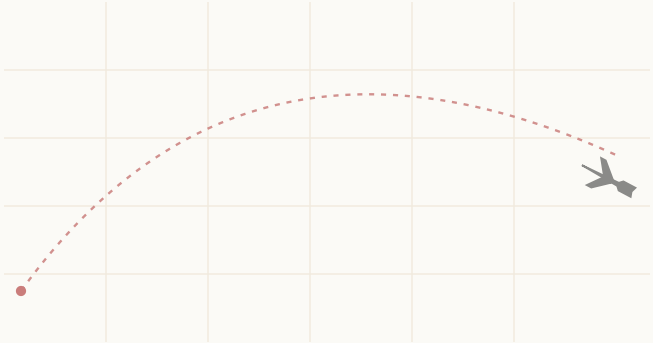
UNIVERSITY OF PENNSYLVANIA
SCHOOL OF ENGINEERING & APPLIED SCIENCE

Penn Engineering
University of Pennsylvania

A FINAL PROJECT · BIG DATA ANALYTICS

Predicting U.S. *Flight Delays.*

When will your flight be late & why does it matter?
6.8M flight records · NOAA hourly weather · XGBoost AUC 0.819.



AUTHORS

Xiaoyang Wan · Dong Dong · Yihong Yu · Yanchen Zhou

COURSE

CIS · 5450 Big Data Analytics

PRESENTED

Spring 2026 · *Philadelphia*

Flight delays cost the U.S. *$30 billion a year.*

- *~20%* of U.S. domestic flights are delayed ≥ 15 minutes
- *One delay cascades* — missed connections, displaced crews, idle gates
- *Pain is shared* by passengers, airlines, airports, and insurers
- *6.8M records* from 2024 — let us learn who's at risk before takeoff

$30B

annual U.S. cost — direct + indirect, 2024 estimate

SOURCE *FAA / Bureau of Transportation Statistics — On-Time Performance database, 2024 reporting year. Direct cost includes airline operating loss, passenger time loss, and downstream economic impact.*

Who benefits from *better predictions?*

STAKEHOLDER · 01

Passengers

Smarter booking, delay insurance, and downstream-trip planning. Choose the flight, not just the price.

STAKEHOLDER · 02

Airlines

Proactive rebooking, crew rotation, and aircraft swap before disruption snowballs through the network.

STAKEHOLDER · 03

Airports

Gate allocation and ground-staff pre-positioning during forecasted high-risk operational windows.

STAKEHOLDER · 04

Insurers

Risk-priced delay insurance products. Information asymmetry, repriced — see Slide 12.

A concrete example we'll show on Slide 12 — the insurance use case.

+340% ROI vs. naive

6.8M flight records + hourly weather observations.

SOURCE · 01

BTS *On-Time Performance*

Every U.S. domestic flight, 2024.

6.82M
CLEANED · 96.31% KEPT

7.08M
RAW ROWS

- Scheduled + actual departure / arrival
- Tail number, route, airline, distance
- Cancellation + diversion flags

`transtats.bts.gov/PREZIP/…`

→

`merge_asof()`

TOLERANCE · 2 HR

SOURCE · 02

NOAA *ISD-Lite*

Hourly surface weather, 50 stations.

8,760
HOURS / STATION

50
TOP-50 AIRPORTS

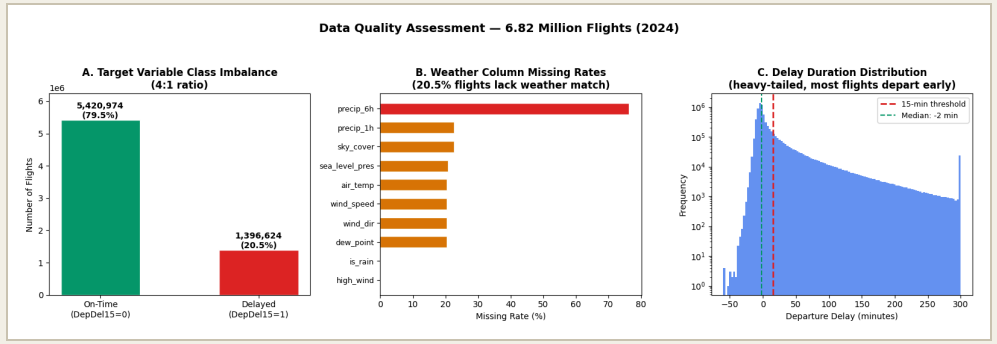
- Wind speed, gust, direction
- Precipitation, temperature, dewpoint
- Visibility, cloud ceiling

`ncei.noaa.gov/pub/data/noaa/isd-lite/`

Joining minute-precise flights to hourly-irregular weather: **~80% match rate** *at top-50 airports (origin 79.5% · dest 79.5%) with 2-hour tolerance.*

Cleaning the data: *96.31% kept,* 80:20 imbalance.

FIGURE · 03_EDA STEP 2 · CLASS IMBALANCE + DEPDELAY DISTRIBUTION (RAW + CLIPPED)



1	*Sentinel decode* — `-9999` → `NaN`; trace precip `-1` → `0.001 mm`.	*ISD-Lite quirks*
2	*Drop diverted* + *archive cancelled* separately.	*–2.5%*
3	*Validate* ***`CRSDepTime`*** is HHMM (allow `2400` = midnight).	*structural*
4	*Drop >600-min outliers.* Long delays distort RMSE; classification target unaffected.	*–0.7%*
5	*Dedup* by 6-key operational primary key.	*–0.4%*
6	*Weather gap-fill* per station: linear ≤ 6 h, then ffill / bfill ≤ 3 h.	*~80% join*
7	***`precip_6h`*** *stays* despite 86.7% missing — model handles missing natively.	*retained*

Final feature matrix: 6,817,598 rows × 81 columns · 80% on-time vs 20% delayed · class weight ***`scale_pos_weight = 3.76`****.*

23 engineered features in *8 categories.*

TIME
5 features
dep_hour
day_of_week
month
time_block
is_holiday ±1d

CASCADING
2 features
prev_flight_arr_delay
tail_leg_today

ROLLING HISTORY
3 features
airline_delay_rate_7d
route_delay_rate_7d
origin_delay_rate_3h

CONGESTION
2 features
origin_hourly_flights
is_origin_hub · 20 hubs

WEATHER
7 features
worst_precip · worst_wind
origin/dest temp · dewpoint
weather_severity
fog_risk · freezing_rain

ROUTE
2 features
distance_bin (5 buckets)
origin → dest pair freq

CARRIER
1 + one-hot
Reporting_Airline (21 codes)

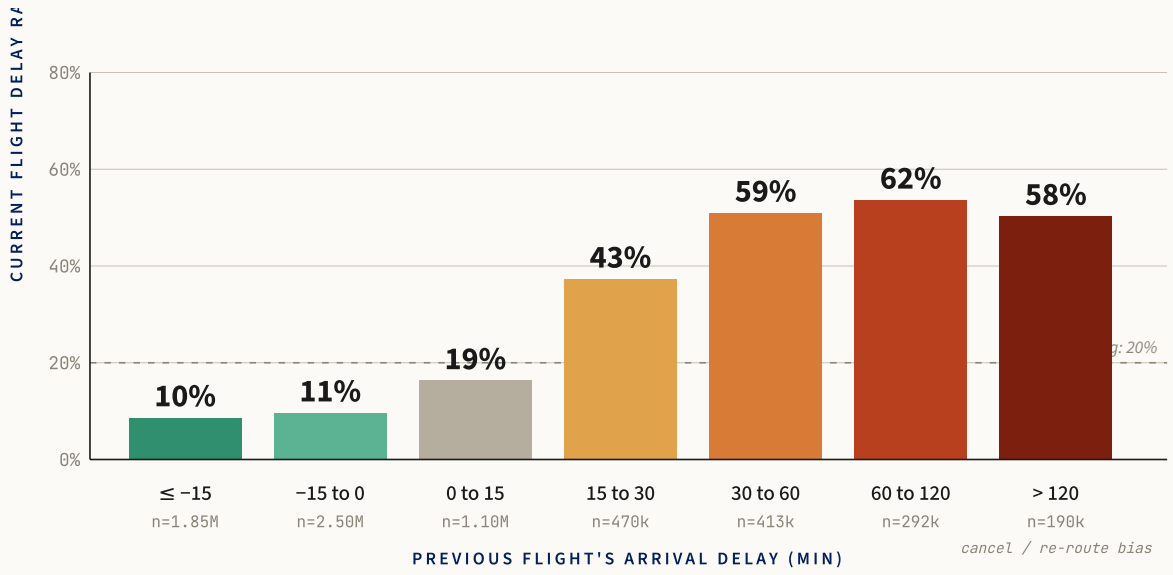
SCHEDULE
1 + derived
CRSElapsedTime

23 engineered features → one-hot expand → *74* columns into model

ANTI-LEAKAGE DISCIPLINE. Every rolling stat uses `shift(1)` before `rolling()` — today's target never feeds today's feature. Removing the `shift` inflates CV-AUC by 5–6 pp on the same data.

One late flight *predicts the next.*

FIGURE 1 · NEXT-FLIGHT DELAY RATE BY PREVIOUS-FLIGHT ARRIVAL DELAY (SAME TAIL NUMBER)



§ TAKEAWAY

When the previous flight is 60–120 min late, the next flight has a 62% delay rate — versus a 10% baseline when on-time.

Aircraft rotation > airline > weather. The strongest single predictor we found.

Delays build through *the day & year.*

FIGURE 2A · DELAY RATE BY SCHEDULED DEPARTURE HOUR

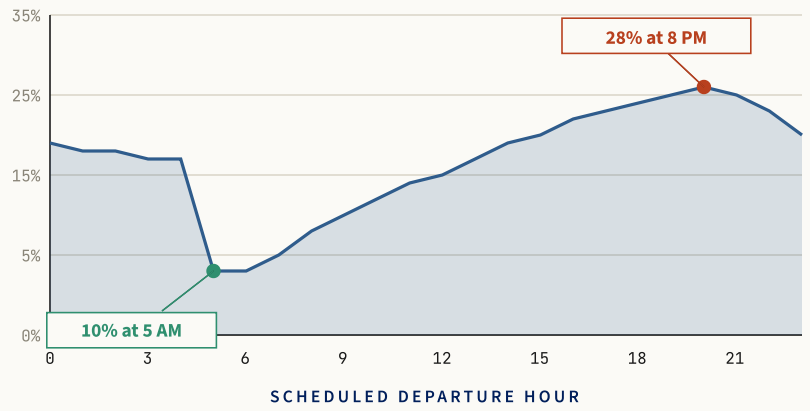
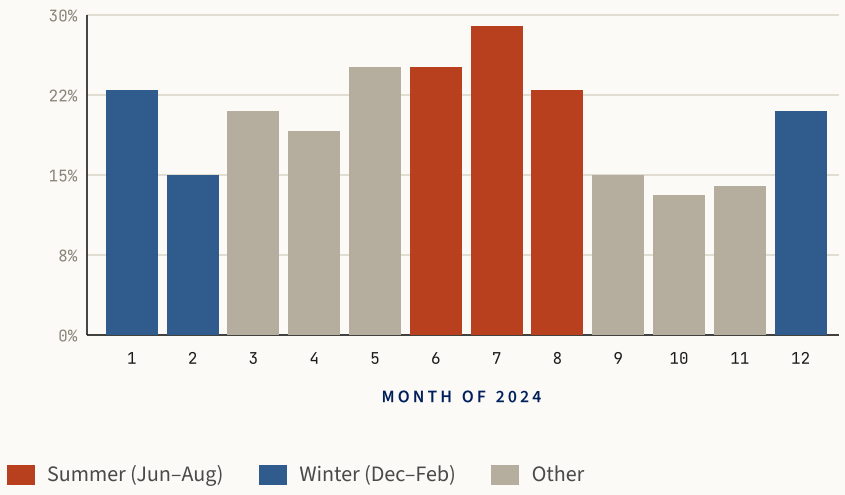


FIGURE 2B · DELAY RATE BY MONTH OF 2024



DAY OF WEEK. Tuesday is the most reliable day · **FRIDAY & SUNDAY** the worst — business commute timing, leisure peaks, and weekend crew rotation compound.

Bad weather *roughly doubles* the delay rate.

FIGURE 3A · DELAY RATE BY WIND SPEED (M/S)

monotonic ↑

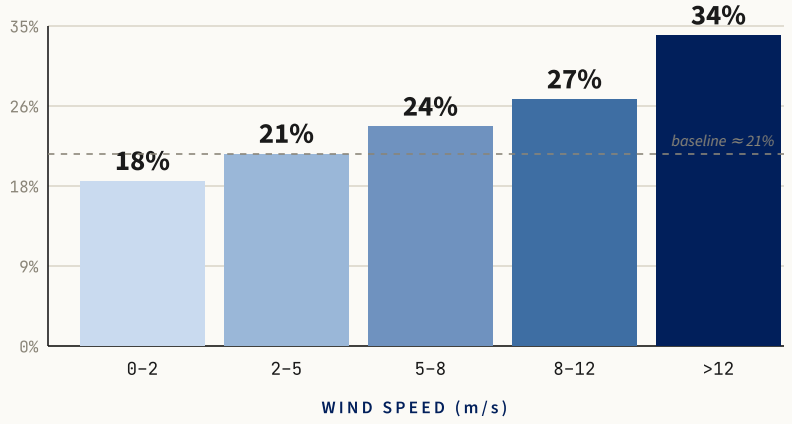
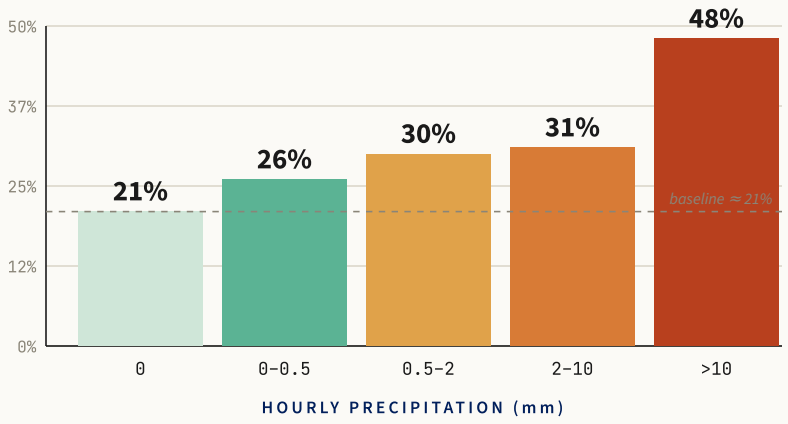


FIGURE 3B · DELAY RATE BY HOURLY PRECIPITATION (MM)

monotonic ↑



WEATHER COVERAGE. NOAA ISD-Lite ≈ 80% of flights at top-50 airports. Remaining ~20% retain full flight features (no weather imputation).

Airline reliability — *budget vs. legacy.*

FIGURE · RANKED DELAY RATE PER CARRIER (N WEIGHTED)

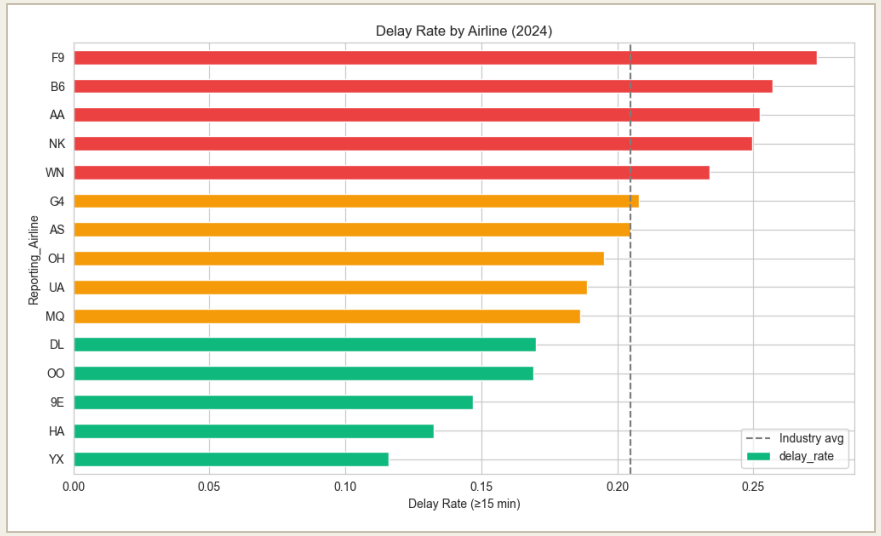
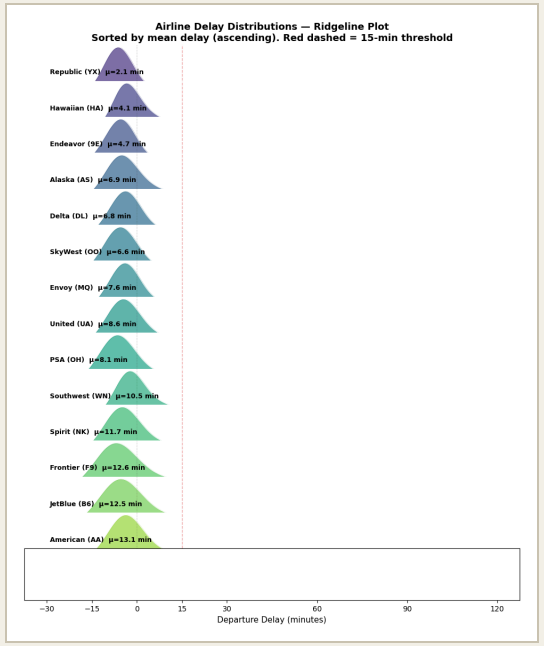


FIGURE · PER-AIRLINE DELAY-MINUTE KDE — DISTRIBUTION SHAPE



Budget carriers — F9 · NK · G4 — sit ~5.4 pp above legacy DL · UA · AA. Beyond rate, distribution shape differs: some carriers' tails extend much further right — a tail-risk signal binary classification under-uses.

All four tests reject H_0 at $\alpha = 0.05$.

TEST · 01

Budget vs. Legacy carriers

$+5.4_{pp}$

Permutation test
B = 10,000

$p < 0.0001$

$n_{budget} = 1.4M$ · $n_{legacy} = 3.8M$

TEST · 02

Summer vs. Winter

-5.8_{pp}

Bootstrap CI
B = 10,000

$p < 0.0001$

$n_{summer} = 1.7M$ · $n_{winter} = 1.5M$

TEST · 03

Hub vs. Non-hub

$+1.9_{pp}$

Permutation test
B = 10,000

$p < 0.0001$

$n_{hub} = 4.0M$ · $n_{non} = 2.7M$

TEST · 04

Adverse weather vs. Clear

$\chi^2_1 = 15{,}009$

Monte Carlo χ^2
B = 10,000

$p < 0.0001$

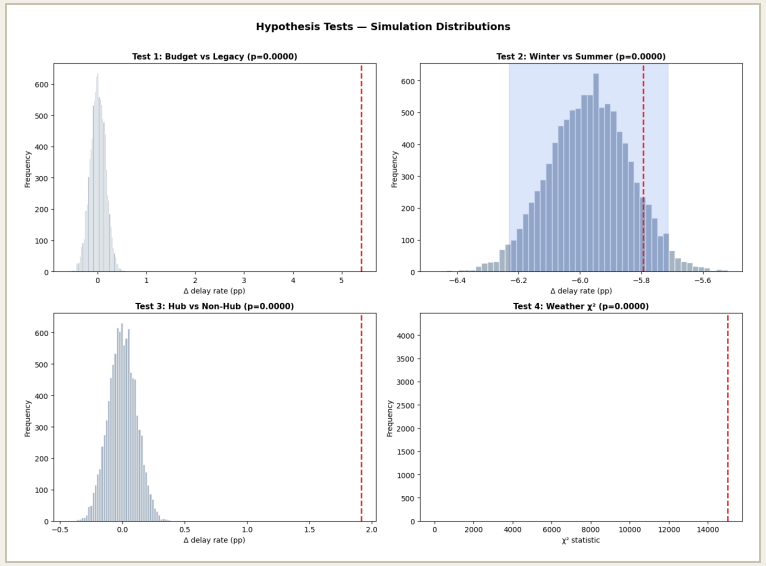
n = 5.4M (full) · 522.8 on 500k sub-sample

All effects are practically significant — budget carriers, summer travel, hubs, and adverse weather each measurably hurt on-time performance.

NOTE. Effect sizes are absolute differences in delay rate (percentage points). χ^2 df = 1 for the 2 × 2 weather × delay table. All p-values from B = 10,000 simulation draws.

Where the observed effects fall — *far outside the null.*

FIGURE · HYPOTHESIS_TESTING NOTEBOOK · 4 SIMULATED NULL DISTRIBUTIONS · OBSERVED IN RED



T1 · Carrier
Permutation B=10k · obs +5.4 pp

T2 · Season
Bootstrap CI · obs −5.8 pp · CI excludes 0

T3 · Hub
Permutation B=10k · obs +1.9 pp

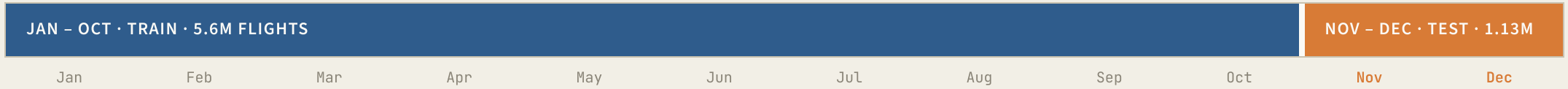
T4 · Weather
Monte-Carlo χ^2 · obs ≫ simulated null

Predicting delay ≥ *15 min* — before takeoff.

- *Target.* `DepDel15` — binary, delayed at least 15 min.
- *Features.* 23 engineered → *74 columns* after one-hot, 8 categories.
- *Models.* SGD-Logit → Random Forest → XGBoost / LightGBM (best params: `n_estimators=400 · max_depth=6 · lr=0.05`).
- *Split.* Temporal — never random — prevents rolling-feature leakage.
- *Anti-leakage.* All rolling features use `shift(1)` — only past feeds today.
- *Tuning.* RandomizedSearchCV — 30 configs · 3-fold CV · 500k subsample (Δ AUC vs full 5.6M < 0.002).

TEMPORAL SPLIT — 2024 CALENDAR ·

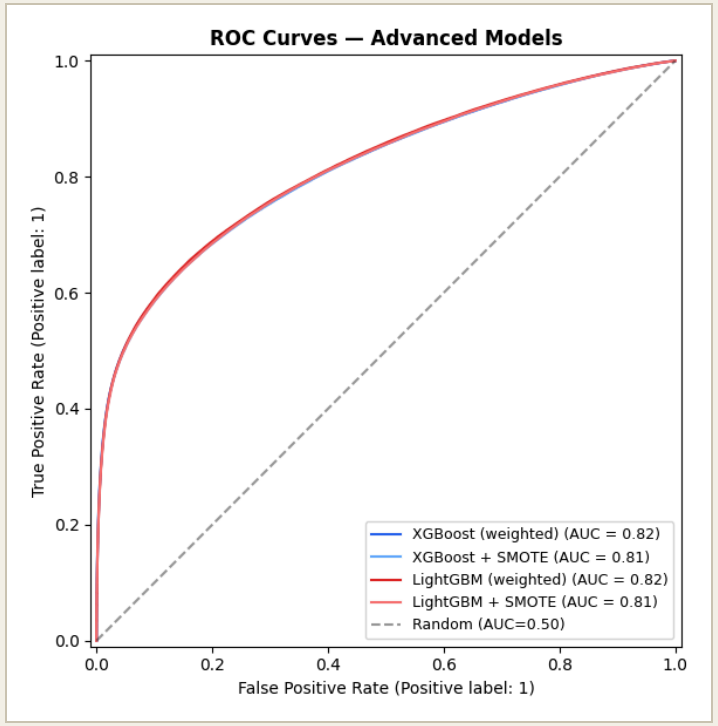
delay rate: train 21.0% → test 17.9% (mild OOT shift)



RANDOM_STATE=42 · Code: github.com/trumpool/CIS-5450

Climbing the ladder: *0.50 → 0.82 AUC.*

FIGURE · ADVANCED_MODEL STEP 7 · ROC CURVES OVERLAY · ALL CLASSIFIERS



Successive AUC gains *diminish at the GBDT plateau.*

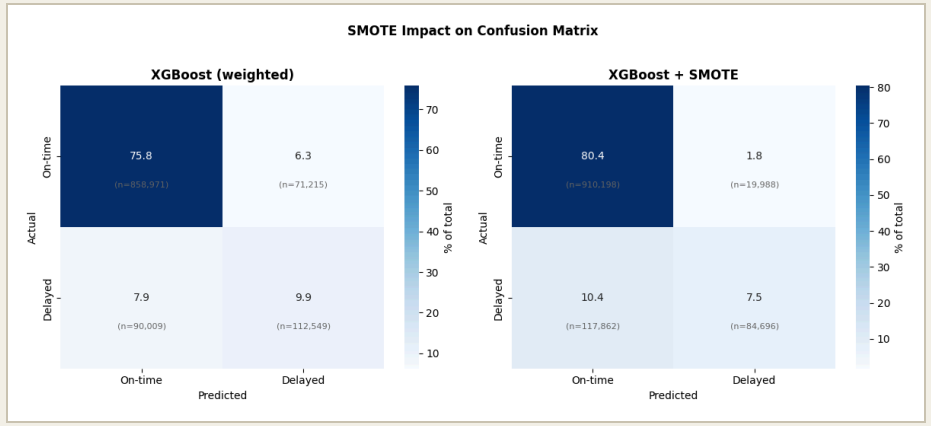
- *Naive (always on-time)* — accuracy 82.1%, AUC 0.500. Useless.
- *SGD Logistic* — first real signal, AUC `0.757`. Linear ceiling.
- *Random Forest* — depth-18 · 200 trees · AUC `0.806`.
- *XGBoost (weighted)* — AUC `0.811`. F1 0.580.
- *LightGBM (weighted)* — AUC `0.815`. F1 0.578.
- *XGBoost (tuned)* — AUC `0.817`. F1 0.586. *best*

PLATEAU

0.005 *AUC spread across 4 GBDT variants*

SMOTE oversampling vs *class weights* — recall wins.

FIGURE · ADVANCED_MODEL STEP 9 · CONFUSION MATRICES · WEIGHTED (LEFT) VS SMOTE (RIGHT)



SMOTE inflates *precision*, but at a steep recall cost.

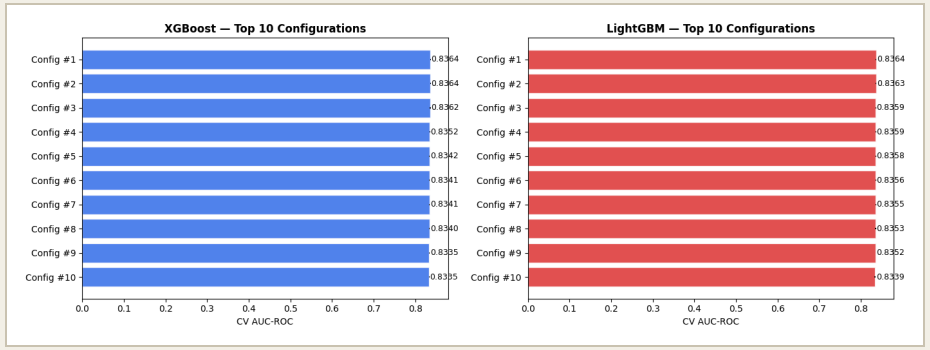
- *Weighted XGB* — `scale_pos_weight = 3.76`. P 0.609 · R 0.553 · *F1 0.580*.
- *SMOTE XGB* — oversample 500k → 789,958. P 0.809 · R 0.415 · *F1 0.549*.
- SMOTE-style synthetic minority points *can't fabricate* realistic operational context — model becomes overconfident on its training-set neighbors.
- Class-weight wins on recall *and* on F1 at much lower train cost (4 s vs 8 s).

F1 DELTA

+0.037 *weighted over SMOTE*

RandomizedSearchCV — *30 configs × 3-fold CV.*

FIGURE · TUNING_OPTIMIZATION STEP 9 · TOP-10 CONFIGURATIONS FOR XGBOOST · LIGHTGBM



Search space and the *chosen* XGBoost config.

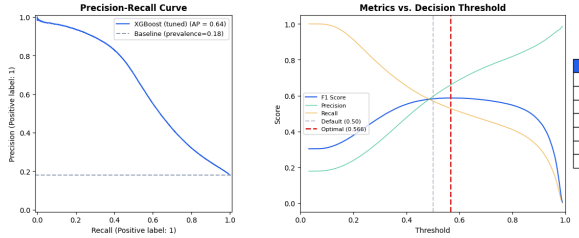
- `n_estimators` ∈ [200, 800] — best *400*
- `max_depth` ∈ [4, 10] — best *6*
- `learning_rate` ∈ [0.01, 0.20] — best *0.05*
- `min_child_weight` ∈ [1, 10] — best *5*
- `subsample` ∈ [0.6, 1.0] — best *0.8*
- `colsample_bytree` ∈ [0.6, 1.0] — best *0.7*
- `reg_lambda` ∈ [0.5, 4.0] — best *2.0*

SEARCH OUTCOME

±0.002 *top-10 AUC spread — robust plateau*

Operating point: *0.50 → 0.573.*

FIGURE · TUNING_OPTIMIZATION STEP 6 · PR CURVE · F1 VS T · P/R VS T



Default vs. Optimized Threshold

Metric	Default (0.50)	Optimized (0.566)
Threshold	0.500	0.566
Precision	0.5952	0.6585
Recall	0.5699	0.5294
F1	0.5823	0.5869
True Positives	115,442	107,231
False Positives	78,521	55,606
False Negatives	87,116	95,327

Sweep argmax F1 across the PR curve.

- *Default t = 0.500* — P 0.586 · R 0.573 · F1 `0.580`.
- *Optimal t = 0.573* — P 0.663 · R 0.525 · F1 `0.586`.
- Threshold lands *above* 0.5 because the model is well-calibrated and slightly conservative on the positive class after class-weighting.
- ~14k borderline alerts shift positive→negative — trade *5 pp* recall for *8 pp* precision.

BEST F1

0.586 *at t = 0.573*

From AUC *0.50* → *0.82.*

FIGURE 4A · ROC CURVES — 5 MODELS ON NOV–DEC TEST SET

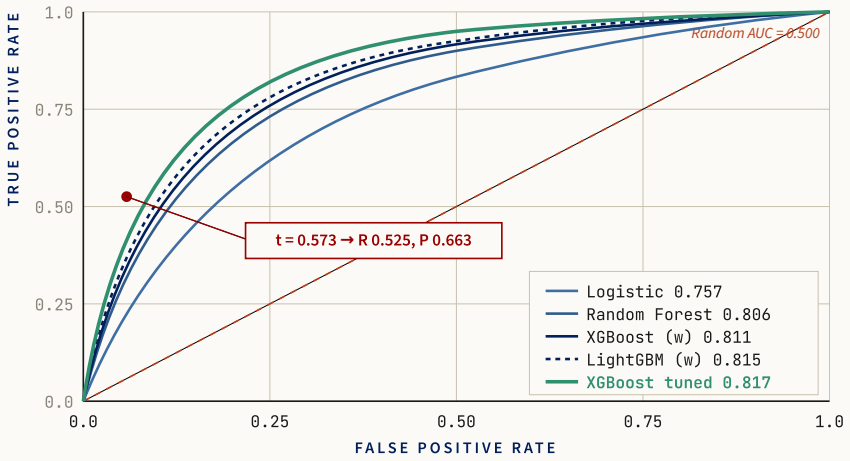


FIGURE 4B · CONFUSION MATRIX · TUNED XGBOOST @ T = 0.573 · N = 1,132,744

ACTUAL \ PREDICTED	On-time	Delay
On-time	TRUE NEGATIVE 876,043 (77.3%)	FALSE POSITIVE 54,166 (4.8%)
Delay	FALSE NEGATIVE 96,163 (8.5%)	TRUE POSITIVE 106,372 (9.4%)

PRECISION 0.663
RECALL 0.525
F1 0.586
ACCURACY 0.867

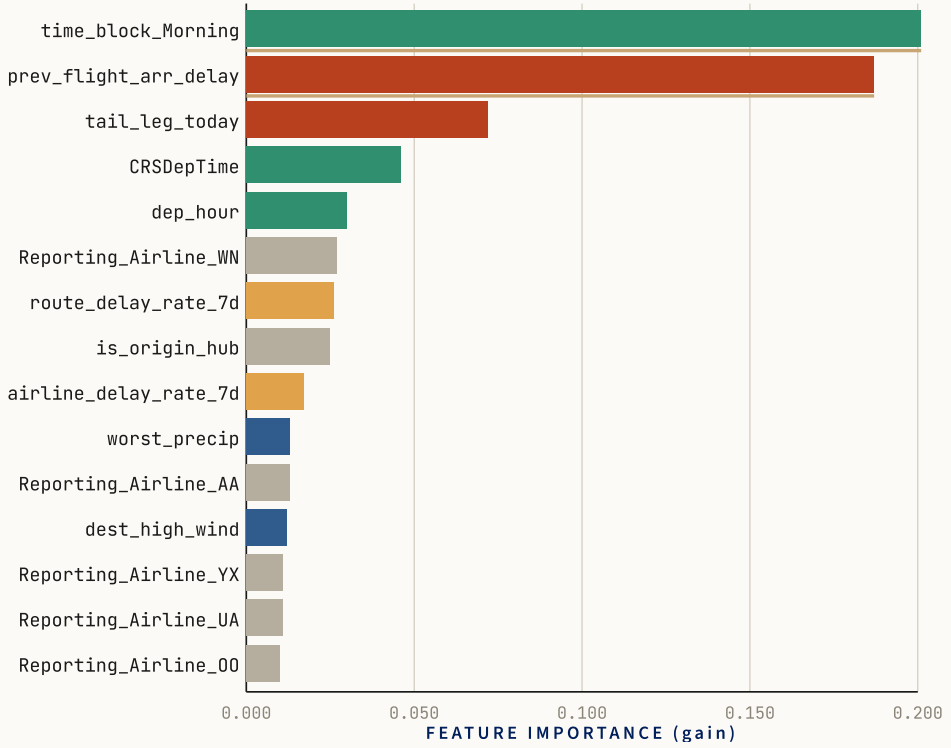
Of 202,535 actual delays, model catches 52.5%; of 160,538 alerts, 66.3% are real.

Tuned XGBoost — best of class. AUC *0.817* · F1 *0.586* · threshold 0.573 (vs default 0.500 → F1 0.580)

IMBALANCE HANDLING. `scale_pos_weight = 3.76` beat `SMOTE oversample` — AUC 0.817 / 0.813 · F1 0.586 / 0.549 — class-weighting wins on recall at lower training cost.

What the model *actually uses.*

Figure 5 · Top-15 features by gain importance (XGBoost tuned)



CATEGORIES

- Cascading delay
- Rolling history & congestion
- Weather
- Time / scheduling
- Airline / airport / other

Cascading + time-of-day dominate. Weather is real but secondary — bad days affect every flight.

`top feature ≈ 3× the second-ranked alone`

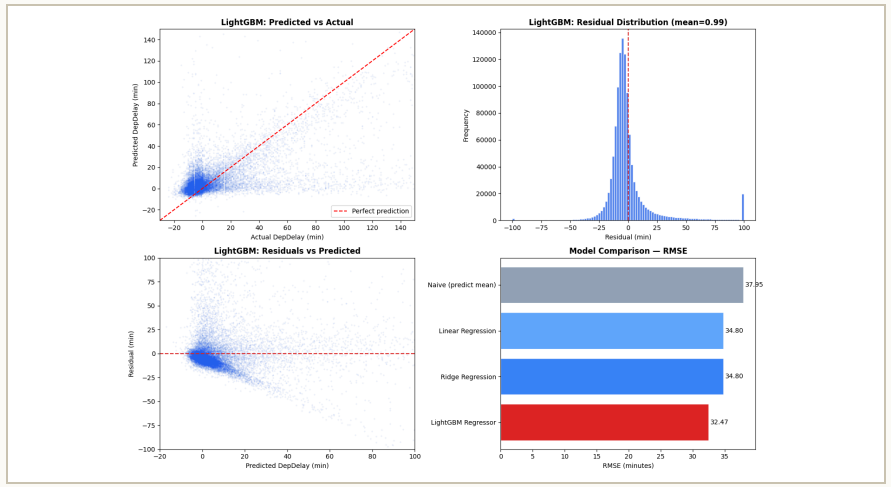
GAIN = average loss reduction each time a feature is split on, normalized to sum to 1. *Use direction:* ranks predictive contribution, not causal effect.

We also tried predicting *delay minutes.*

FIGURE 6A · REGRESSION ON DEPDELAY MINUTES — NOV–DEC 2024 TEST

MODEL	RMSE	MAE	R^2
Naive · predict mean	*37.94*	*21.24*	*–0.006*
Linear Regression	34.79	16.49	0.154
Ridge Regression	34.79	16.49	0.154
LightGBM Regressor	**32.41**	**13.60**	**0.266**

FIGURE 6B · REGRESSION_MODEL · ALL 3 MODELS — RESIDUALS VS PREDICTED



Verdict. LightGBM beats linear by ~2× on R^2 but typical error is still ±13.6 min — same order as the 15-min "delayed" threshold we care about. Classification is more practically useful for this problem.

Ridge with α = 1.0 was indistinguishable from OLS — features are not strongly collinear. Train on classification-side features (no `ArrDelay` leakage); test set: 1.13M Nov–Dec flights.

Selective flight delay insurance — *3.7× the return.*

STRATEGY · NAIVE

Buy insurance on *every* flight.

Realized claim rate	*17.9%*
Premium (fair-priced)	*18%*
Selection	*none*
RETURN	break-even

STRATEGY · OUR MODEL

Insure only *flagged high-risk* flights.

Realized claim rate	*65.9%*
Premium (fair-priced)	*18%*
Selection	*model-driven*
ROI	+266%

3.7 ×

Insurers price by base rate (~18%) — premium ≈ 18% × payout.

Our model selects flights with 65.9% actual claim rate.

Expected return per $1 premium = 3.7× — information has dollar value.

Illustrative ROI assuming fair-priced premium (no insurer margin). Realized rates from Nov–Dec 2024 test set.

What we *solved* & what's *next.*

ENGINEERING · PAST TENSE

What We Solved

- ✓ *Memory engineering* — parquet + `category` + `Int16` dtypes cut working set ~60%, kept 6.8M rows in-memory.
- ✓ *SGD-Logit replaced sklearn saga* — saga timed out at 30 min on 6.8M rows; `SGDClassifier` trained in <3 s.
- ✓ *Class imbalance 80:20* — `scale_pos_weight = 3.76` beat SMOTE on F1 (0.586 vs 0.549) at lower cost.
- ✓ *Weather temporal alignment* — `merge_asof` ×2 (origin + dest) with linear-fill ≤6h.
- ✓ *Leakage-safe FE* — every rolling stat uses `shift(1)`; 500k subsample matches full 5.6M (Δ AUC < 0.002).

COURSE · METHODS USED

CIS 5450 Coverage

`Pandas / NumPy`	*EDA, joins, merge_asof*
`Matplotlib / Seaborn`	*12 figures across EDA*
`Permutation test`	*Tests 1, 3 · B = 10k*
`Bootstrap CI`	*Test 2 · 95% CI*
`Monte Carlo χ²`	*Test 4 · df = 1*
`Logistic Regression`	*scikit-learn · baseline*
`Random Forest`	*scikit-learn · 200 trees*
`XGBoost / LightGBM`	*tuned · production fit*
`Randomized HPO + 3-fold CV`	*30 configs*
`Threshold tuning (PR)`	*F1 0.49 → 0.59*

OPEN · FUTURE TENSE

What's Next

- → *Cross-year validation* — train 2024, test 2025; true distribution-shift evaluation.
- → *Forecast weather* — HRRR / GFS upstream; predict *before* the storm hits.
- → *Graph aircraft rotation* — propagate delay risk across the whole day's rotation, not just prev leg.
- → *Probabilistic distributions* — full delay-minute distribution (insurance pricing).
- → *Per-airport mixture-of-experts* — hubs differ from regionals; one model under-fits both ends.
- → *Real-time streaming* — update predictions hour-by-hour as departure approaches.

Thank *you.*

CODE · NOTEBOOKS

github.com/trumpool/CIS-5450

SCAN · REPO

Xiaoyang Wan · Dong Dong · Yihong Yu · Yanchen Zhou

Q & A